

# FactorJEPA: World Models That Predict Crowded, Chaotic City Streets

Anonymous submission

## Abstract

World models aim to capture and predict the structure and dynamics of the physical world, and Joint Embedding Predictive Architectures (JEPA) are a compelling direction. We study a largely unexplored regime: populous, crowded, and chaotic Global South urban environments, which we call **DENSE-WORLD**, where soft spatial boundaries, extreme agent heterogeneity, persistent occlusion, and rapid social negotiation replace the lane-structured assumptions behind existing evaluations. We release DENSEWORLD 1.0: 115,687 videos (276 hours) of drive-through, walk-through, and aerial footage, each 4 to 10 seconds, spanning 22 Indian cities and 714 long-form sources, of which 86,831 carry automatically derived layout, agent, and interaction targets. Ten off-the-shelf frozen encoders (DINOv2, I-JEPA, LeJEPA, and V-JEPA 1/2.0/2.1) all collapse into a narrow 37.5% to 44.4% band on an 11-way motion-class probe with a 19.5% majority baseline, whereas DENSEWORLD-adapted encoders reach 50.3% to 53.2%.

We introduce **FactorJEPA**, which makes world structure a first-class predictive primitive. Rather than adapting a JEPA predictor on undifferentiated video, it performs *predictor surgery*: a three-stage curriculum over automatically separated layout, agent, and interaction views, updating at most 8/48 (2B) and 6/40 (1B) encoder blocks with the predictor. Trained on 86,831 factor-annotated videos and evaluated on 23,106 held-out samples (1B V-JEPA 2.1), it cuts future-latent L1 by 10.6%, causal future-block L1 by 8.9% and mask-ratio slope by 27.4% against the frozen backbone, and raises motion-cosine separation to  $15.3\times$  the frozen value, each gain exceeding  $70\times$  the paired 95% confidence interval. On both prediction diagnostics it separates in its favour against all six adaptation techniques we evaluate, namely full fine-tuning, linear probing then fine-tuning (LP-FT), low-rank adaptation (LoRA), weight-decomposed low-rank adaptation (DoRA), automatic relative-gradient-norm block selection (Auto-RGN), and vanilla continual self-supervised learning (continual-SSL), at both backbone scales, giving 24 of 24 paired wins, and beats the strongest full-scale baseline, LoRA, by +2.8%, +1.6%, +25.2% and +17.8%.

20-arm (2B) and 14-arm (1B) studies show rankings replicate across backbone scale at  $\rho = 0.895$  to  $0.978$ , establishing the 1B model as a half-cost proxy for 2B selection. On a 10k screening subset, factorization concedes maximally linear

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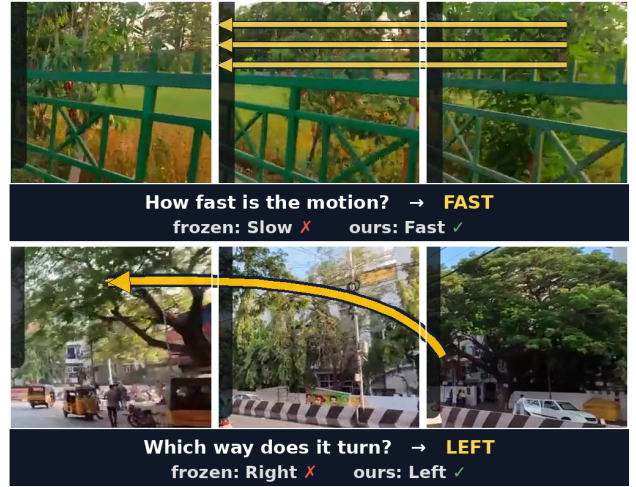


Figure 1: **Same question, two encoders.** Each row is a three-frame filmstrip of one held-out clip posed as a multiple-choice question, answered from an identical probe head over the frozen V-JEPA 2.1 encoder vs. ours (factor-view predictor surgery); only the backbone differs. *Top*: motion speed (ours 69.8% vs frozen 60.9%). *Bottom*: turn direction.

motion readout to full fine-tuning ( $-44.9\%$  at 2B,  $-33.4\%$  at 1B), a trade-off that disappears at full scale, where FactorJEPA leads motion by +25.2% and gains more than LoRA on all four diagnostics. We release the dataset, checkpoints and the complete 15-diagnostic evaluation harness.

## Introduction

World models, which capture the structure and dynamics of the physical world, are emerging as a foundation for agents, robotics, and physical AI. Joint Embedding Predictive Architectures (JEPA) are a compelling direction: they predict masked or future content directly in embedding space, avoiding costly pixel generation while promoting semantic abstraction. Yet the settings on which such models are evaluated are overwhelmingly lower-density and lane-structured, far from the environments where physical AI must ultimately operate.

We study a largely unexplored regime: populous, crowded, and chaotic Global South city streets, which we call **DENSE-**

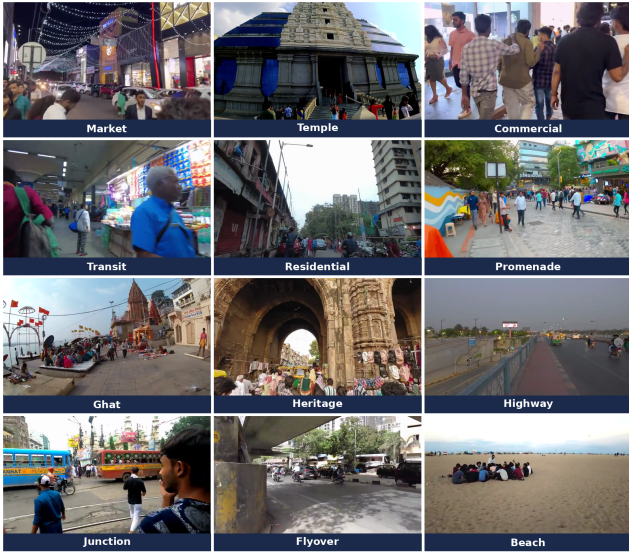


Figure 2: **DENSEWORLD 1.0 scene-type coverage.** Twelve representative scene types (market, temple, commercial, transit, residential, promenade, ghat, heritage, highway, junction, flyover, beach) across 22 Indian cities, shown from ground-level pedestrian viewpoints.

**WORLD** (Figure 2). Here, soft spatial boundaries, extreme agent heterogeneity, persistent occlusion, and rapid social negotiation under mixed traffic replace the lane-structured assumptions behind existing benchmarks. These are precisely the properties a world model must respect to forecast how a scene will evolve, and precisely the ones that off-the-shelf encoders are not trained to preserve.

Frozen JEPA encoders struggle in this regime: they fail to preserve dense interaction dynamics under heterogeneity and partial observability, and generic fine-tuning closes only part of the gap. The difficulty is structural rather than a matter of scale. The future of a crowded street is governed by *who* is where (agents), *what* the scene affords (layout), and *how* agents negotiate (interactions); a monolithic latent blurs these together and lets the predictor take cross-factor shortcuts.

Our central idea is to make world structure a *first-class predictive primitive*. Rather than adapting a JEPA predictor on undifferentiated video, **FactorJEPA** performs *predictor surgery*: a three-stage curriculum over automatically separated *layout*, *agent*, and *interaction* views, with a visibility gate that preserves partially observed agents and separated subspaces that discourage cross-factor shortcuts. Figure 1 illustrates the payoff on held-out clips: the same probe question is answered more accurately by our factor-view predictor surgery than by the frozen backbone, with only the encoder changed.

We make three contributions. First, we release **DENSEWORLD 1.0**: 115,687 videos (276 hours) across 22 Indian cities, 86,831 of them with automatically derived layout, agent, and interaction targets. Second, we introduce **FactorJEPA** and predictor surgery, which turn factorized world structure into an explicit training signal for the predictor rather than a monolithic fine-tuning objective. Third, on a 15-

diagnostic evaluation harness, FactorJEPA separates from all six fine-tuning baselines (24 of 24 paired wins) and its rankings replicate across backbone scale ( $\rho = 0.895$  to 0.978), making the 1B model a half-cost proxy for 2B.

## DENSEWORLD: A Benchmark for Populous, Crowded, and Chaotic Global South

We use DENSEWORLD to denote a critical but underrepresented world-modeling regime: populous, crowded, and chaotic Global South urban scenes. It is defined not only by geography, but by measurable properties: high agent count density, high agent occupancy, persistent occlusion, agent heterogeneity, and interaction pressure. This regime matters because many emerging urban environments combine high population density, mixed mobility systems, diverse infrastructure patterns, and rapid spatial change, making them scientifically important yet underrepresented in world-model benchmarks (Mahendra et al. 2021; Ellis and Roberts 2016). Thus, world models must handle uncertainty, density, and relational complexity that is less prominent in lower-density, lane-structured benchmarks. See Figure 2 for representative DENSEWORLD scenes.

We characterize DENSEWORLD through four coupled properties. First, it exhibits soft spatial boundaries: road edges, drivable space, and pedestrian regions are often negotiated rather than crisply marked (Varma et al. 2019; Dokania et al. 2023). Second, it contains extreme agent heterogeneity, with pedestrians, cars, two-wheelers, carts, animals, delivery vehicles, and rickshaws coexist locally (Khan and Maini 1999; Asaithambi et al. 2012). Third, it has persistent occlusion from density, clutter, and near-field encounters, producing severe partial observability (Varma et al. 2019; Dokania et al. 2023); prediction must rely on *memory*, *object permanence*, and *relational inference*, not only visible appearance. Fourth, motion is governed less by rigid rule compliance than by rapid social negotiation under mixed traffic and weak lane discipline (Papathanasopoulou and Antoniou 2018; Asaithambi et al. 2012).

Together, these properties make DENSEWORLD a natural stress test for world models: success requires representations that preserve layout, agent state, and interaction dynamics under uncertainty, rather than collapsing into one entangled predictive latent. As detailed below, we measure this regime through agent count density, agent occupancy, occlusion pressure, interaction pressure, and agent heterogeneity. This motivates a predictor that separates layout, entities, and interactions.

### DENSEWORLD 1.0: the Benchmark

We collected 714 long-form YouTube videos at 480p, totalling 276 hours of urban footage across 22 Tier-1 and Tier-2 cities in India. The data spans drive-through, walk-through, and aerial viewpoints, and long-form recordings are segmented using scene-detection and shot-boundary methods into 115,687 short videos of 4 to 10 seconds, of which 86,831 carry automatically derived layout, agent, and interaction targets. DENSEWORLD 1.0 covers commercial, residential, transit, heritage, coastal, and high-density street settings, in-

cluding markets, commercial streets, transit corridors, junctions, flyovers, bazaars, ghats, beaches, and skylines. It further spans variations in the time of day, weather, crowd density, traffic mix, pedestrian-vehicle separation, road geometry, surface quality, encroachment, and lighting. Beyond standard urban actors, the dataset includes locally frequent agents and objects such as auto-rickshaws, cycle rickshaws, street vendors, and animals. Before release, all footage is processed through privacy-preserving filters, including face and license-plate blurring and removal of sensitive segments.

A defining feature of DENSEWORLD is heterogeneous mixed traffic. In many developing-country cities, multiple vehicle and actor types share the same right of way under weak lane discipline, with strong lateral and longitudinal coupling (Khan and Maini 1999; Papathanasopoulou and Antoniou 2018; Asaithambi et al. 2012). This is not a mild variant of standard traffic, but a qualitatively different operating regime. Prior work highlights the coexistence of cars, buses, trucks, auto-rickshaws, scooters, motorcycles, bicycles, carts, and other slow- and fast-moving agents within the same corridor (Khan and Maini 1999; Asaithambi et al. 2012). Here, scene structure does not decompose into *static lanes*, *isolated actors*, and *simple pairwise relations*; spatial support is fluid, visibility is unstable, and behavior is shaped by dense local negotiation.

### How Dense is DENSEWORLD? Comparison with Standard Driving Benchmarks

To make this regime measurable, we compare DENSEWORLD against BDD100K and nuScenes, two widely used benchmarks for autonomous-driving perception and scene understanding (Yu et al. 2020; Caesar et al. 2020). The goal is not to claim that DENSEWORLD scenes merely look different, but to test whether they occupy a quantitatively distinct operating regime for world-model evaluation.

We measure this regime along five axes: agent count density, agent occupancy, occlusion pressure, interaction pressure, and agent heterogeneity. These metrics capture, respectively, the number of dynamic agents, the fraction of image area they occupy, the frequency of partial or heavy occlusion, the density of spatially proximate agent pairs, and the diversity of co-occurring actor categories. Together, they quantify the multi-agent load, visual congestion, visibility degrada-

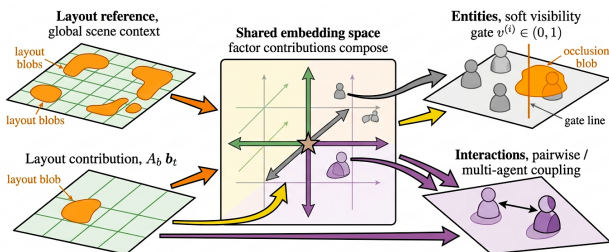


Figure 3: **Factorized predictive channels.** Context is split into *layout*, visibility-gated *agent*, and sparse *interaction* coordinates, then recomposed in the future JEPa embedding, limiting cross-factor leakage.

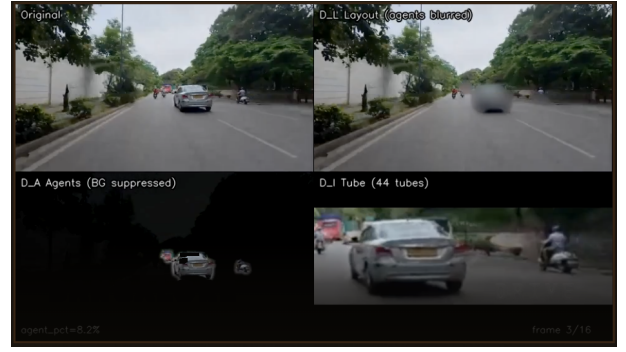


Figure 4: **DINO-based factor-target construction.** Per clip, the pipeline derives *layout*  $D_L$ , *agent*  $D_A$ , and *interaction*  $D_I$  supervision by masking foreground/background and linking agent tubes.

tion, local interaction structure, and traffic diversity of each scene.

For an auditable comparison, all statistics are computed after mapping labels to a shared dynamic-agent taxonomy and matching comparable scene categories across datasets. DENSEWORLD has consistently higher density and occupancy across matched categories, with the largest gaps in market, commercial, and flyover/underpass scenes (quantified in the supplementary material). These results support the central premise of the benchmark: DENSEWORLD is not merely a geographic extension of existing driving data; it defines a higher-density, higher-occlusion, and higher-interaction regime for evaluating predictive world models. Full automatic-annotation details, including thresholds, annotation taxonomy, export format, and loss mapping, are provided in Appendix C.

### Can Fine-Tuning Close the DENSEWORLD Gap?

Before introducing FactorJEPa, we ask whether conventional adaptation can recover the predictive structure required by DENSEWORLD without reorganizing the JEPa predictor. We evaluate three complementary strategies: LoRA (Hu et al. 2022), DoRA (Liu et al. 2024b), and Auto-RGN (Lee et al. 2023). All use the same raw clips, optimization steps, optimizer, masking policy, and evaluation protocol; only the parameter-update mechanism changes.

Let  $x$  denote a video clip;  $M_c$  and  $M_t$  the context and target masks;  $f_\theta$  the online encoder;  $\bar{f}_\theta$  the momentum target encoder; and  $g_\phi$  the predictor. All baselines retain the executed V-JEPa objective (Bardes et al. 2024)  $\mathcal{L}_{\text{JEPa}} = \mathbb{E}_x \|g_\phi(f_\theta(M_c \odot x), M_t) - \text{sg } \bar{f}_\theta(M_t \odot x)\|_2^2$ , where sg is stop-gradient. This data- and step-matched protocol isolates adaptation capacity without introducing a different prediction target or training signal.

The gap is real. On an 11-way motion-class probe (19.5% majority baseline), ten off-the-shelf *frozen* encoders (DINOv2, I-JEPa, LeJEPa, and V-JEPa 1/2.0/2.1) collapse into a narrow 37.5% to 44.4% band (Table 1), whereas



DENSEWORLD-adapted encoders reach 50.3% to 53.2%. Conventional fine-tuning (LoRA, DoRA, Auto-RGN) narrows but does not close the remaining predictive gap, motivating an adaptation that reorganizes *which* factors the predictor attends to rather than only *how many* parameters move.

Formal baseline update rules, protocol-matched configurations, and resource accounting are in Appendix C.

## FactorJEPA: Predictor Surgery over Factorized Views

**Motivation.** With the same notation, a conventional JEPA predictor  $g_\phi$  learns

$$g_\phi : (f_\theta(M_c \odot x), M_t) \mapsto \hat{Y}_{t+\Delta}, \quad Y_{t+\Delta}^* = \text{sg}[\Pi_{M_t} \bar{f}_\theta(x)],$$

by matching  $\hat{Y}_{t+\Delta}$  to  $Y_{t+\Delta}^*$ . The objective specifies *what* to predict, but every clip presents layout, agents and interactions simultaneously, so nothing constrains *which* of them the predictor actually uses.

This ambiguity is consequential in DENSEWORLD, where layout, agent density, visibility and interaction pressure are strongly correlated. A high-capacity predictor can exploit *shortcut mixtures* (crowd texture as a proxy for interaction, visible appearance for object persistence, or road geometry for motion) without recovering the structure governing scene evolution.

FactorJEPA addresses this on the *input* side rather than by redesigning the predictor. We construct three complementary views of every clip, in which layout, agents and interactions are isolated by construction, and adapt a pretrained V-JEPA predictor over them as a staged curriculum. The architecture is unchanged; what changes is the sequence of distributions the predictor is asked to model, and how much of the encoder is free to move at each stage.

### Automatic Factor-View Construction

For each privacy-filtered clip  $x_n$  we run an open-vocabulary detector (Grounding DINO, Swin-B) (Liu et al. 2024a) to localise dynamic agents, and propagate those detections through the clip with a promptable video segmenter (Segment Anything Model, SAM 3) (Carion et al. 2025). Detections are re-seeded at four anchor frames and propagated within each anchor’s segment, which keeps identities stable under the heavy occlusion typical of the regime. This yields, per clip, a set of agent masks and temporally associated tracklets (a mask and box sequence per agent over its lifespan; formal notation in Appendix C). Figure 4 illustrates this factor-target construction.

A deterministic map converts these into three *views* of the same clip, each a video the predictor can be trained on directly (Figure 3):

- $D_L$  (layout): agent regions are suppressed, leaving slowly varying spatial support: road geometry, buildings, drivable area.
- $D_A$  (agents): background context is suppressed, leaving the dynamic actors and their motion.

- $D_I$  (interactions): spatio-temporal tubes around co-occurring, spatially proximate tracklet pairs, isolating local multi-agent coupling.

Checkpoints, prompt vocabulary, association rules, thresholds and per-view coverage are reported in Appendix C.

### Three-Stage Curriculum

Training proceeds over the three views in order of increasing relational complexity,  $D_L \rightarrow D_A \rightarrow D_I$ , with a mixture weight over views at each stage. The ordering is deliberate: layout is the most stationary signal and anchors the encoder before agent appearance and finally interaction structure are introduced.

Encoder capacity is released progressively: at stage  $s$  only the predictor plus the deepest  $K_s = \lfloor \kappa_s L \rfloor$  transformer blocks (of depth  $L$ ) receive gradients, so that at most 8/48 blocks (V-JEPA 2.1 2B) and 6/40 blocks (V-JEPA 2.1 1B) ever move. The lower blocks and the momentum target network follow the frozen / momentum protocol of the underlying V-JEPA implementation.

### Training Objective

FactorJEPA retains the executed V-JEPA objective and adds three auxiliary terms:

$$\mathcal{L} = \mathcal{L}_{\text{JEPA}} + \beta \mathcal{L}_{\text{probe}} + \gamma \mathcal{L}_{\text{motion}} + \lambda \mathcal{L}_{\text{drift}}$$

where  $\mathcal{L}_{\text{JEPA}}$  is the masked future-latent loss above,  $\mathcal{L}_{\text{probe}}$  is a lightweight multi-task read-off used for checkpoint selection,  $\mathcal{L}_{\text{motion}}$  jointly regresses a 13-dimensional optical-flow descriptor and classifies its motion class, and  $\mathcal{L}_{\text{drift}}$  is an  $\ell_2$  anchor to the initialisation  $\theta^{(0)}$  that bounds how far the adapted encoder may travel from the pretrained solution.

### Predictor Surgery

FactorJEPA is initialised from a pretrained V-JEPA model; the online encoder, momentum target encoder, target construction and masking policy are all retained. Because only the top- $K$  blocks and the predictor are updated, and because  $\mathcal{L}_{\text{drift}}$  penalises displacement from  $\theta^{(0)}$ , adaptation is deliberately *local*: it changes how the predictor is trained, not what it is made of. We refer to this as *predictor surgery*. Exact stage boundaries, mixture weights, unfreeze fractions, optimiser settings and measured GPU-hours are reported in Appendix C.

## Experiments & Evaluation

We evaluate whether FactorJEPA improves the predictive structure of V-JEPA rather than only its downstream semantics, using the four diagnostics defined below.

### Evaluator Independence and Held-out Split

To prevent agreement with the pseudo-label generator from being mistaken for improved world modeling, factor targets (Grounding DINO + SAM 3) are constructed only on the training split, and evaluation uses a disjoint held-out split

Table 1: **Frozen encoders collapse.** Ten frozen encoders on the DENSEWORLD motion probe ( $n_{\text{test}} = 1,825$ ,  $\pm 95\%$  BCa CI): Action top-1 (A) stays in a 37.5–44.4% band (19.5% majority), while adapted encoders reach 50.3–53.2%. M = motion-cos, T = taxonomy F1 ( $\uparrow$  better); shade  $\propto$  column rank.

| Frozen encoder   | A (%) | M     | T     | Frozen encoder    | A (%) | M     | T     |
|------------------|-------|-------|-------|-------------------|-------|-------|-------|
| V-JEPA 2.1 (2B)  | 44.4  | 0.009 | 0.793 | I-JEPA ViT-H      | 39.1  | 0.016 | 0.781 |
| V-JEPA 2.1 ViT-L | 44.2  | 0.004 | 0.788 | V-JEPA 2.0 (SSv2) | 38.8  | 0.007 | 0.776 |
| V-JEPA 1 ViT-H   | 40.5  | 0.007 | 0.702 | DINOv2            | 38.5  | 0.016 | 0.816 |
| LeJEPA ViT-L     | 40.1  | 0.014 | 0.740 | V-JEPA 2 ViT-L    | 37.9  | 0.013 | 0.778 |
| V-JEPA 1 ViT-L   | 39.9  | 0.008 | 0.660 | I-JEPA ViT-G/16   | 37.5  | 0.019 | 0.787 |

**FactorJEPA vs. the best competitor, across scale: separation in 95%-CI units (rows sorted best-to-worst by the 115k panel; dashed line = the 1 $\times$  separation threshold)**

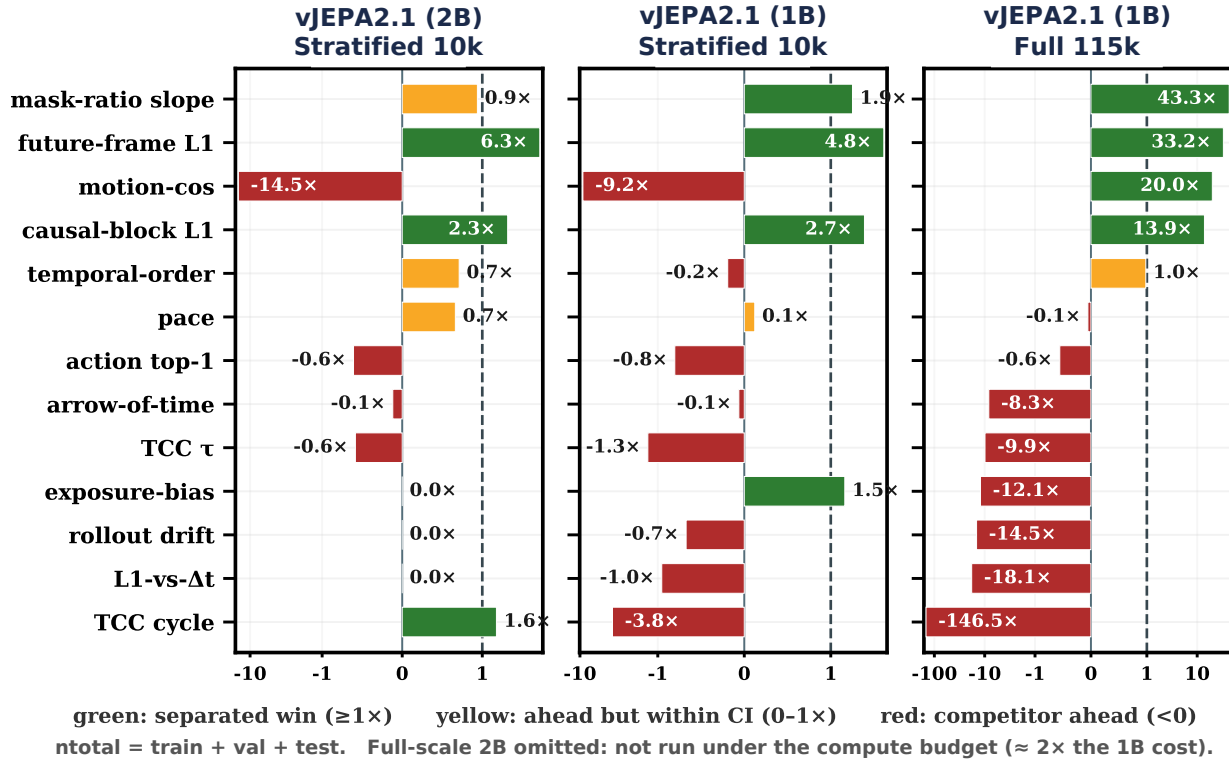


Figure 5: **FactorJEPA vs. the best competitor, across scale.** Per-metric advantage in 95%-CI units at the stratified 10k scale (2B and 1B) and the full 115k scale (1B); green past the dashed 1 $\times$  line = a separated win. Predictive diagnostics separate and *strengthen* at full scale, where the motion-cosine trade-off also reverses to a win. Full-scale 2B was not run (compute budget).

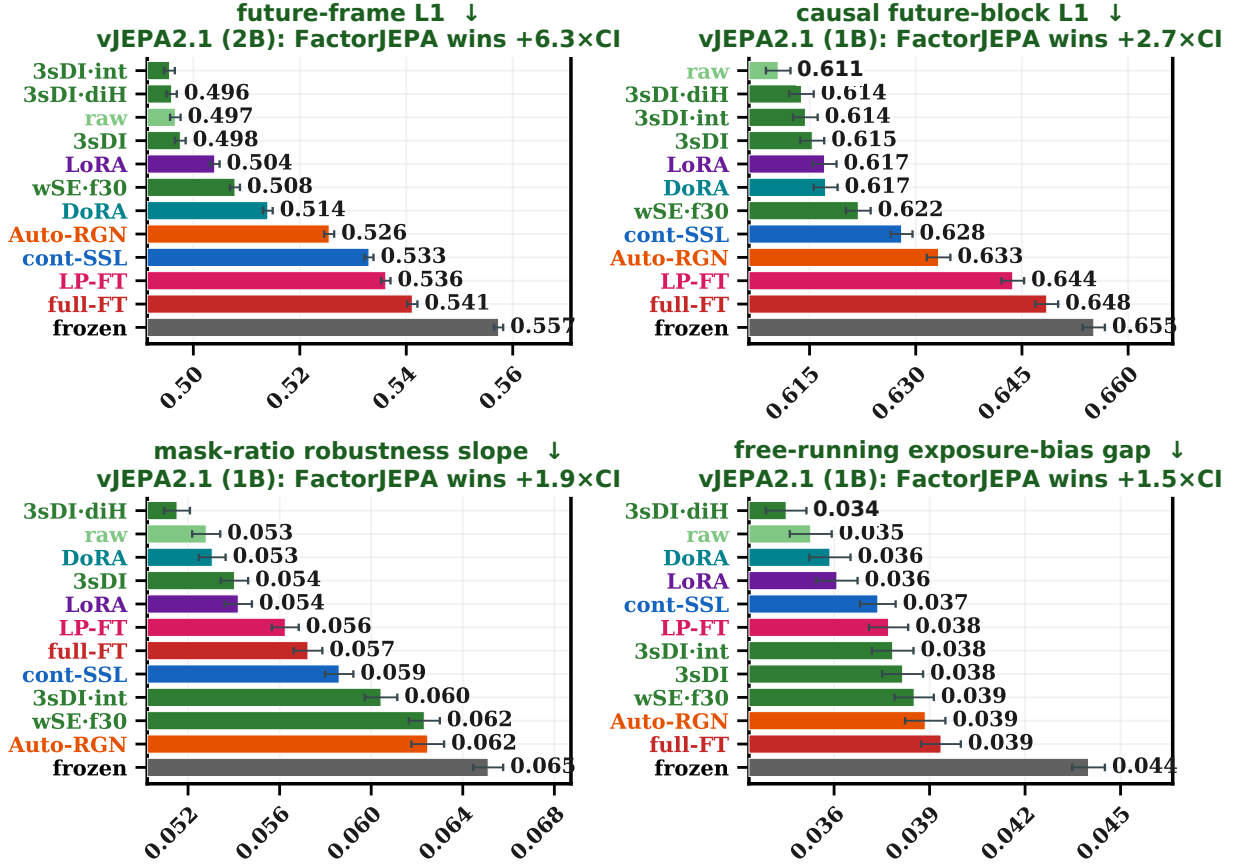
never seen in training, selection, or tuning. No headline metric reuses FactorJEPA’s pseudo-label generator as its evaluator; each is scored by a frozen, independent evaluator (full provenance in Appendix E).

### Protocol and Comparisons

We evaluate V-JEPA 2.1 (Mur-Labadia et al. 2026; Assran et al. 2025) at two scales: V-JEPA 2.1 (2B) and V-JEPA 2.1 (1B), approximately 2B and 1B parameters respectively. The comparison spans the frozen backbone, conventional full fine-tuning, LP-FT (Kumar et al. 2022), parameter-

efficient LoRA, DoRA, Auto-RGN, FactorJEPA-RAW, and full FactorJEPA. FactorJEPA-RAW applies the factorized predictor to raw clips without factor-target supervision, so the chain generic adaptation (LoRA, DoRA, Auto-RGN)  $\rightarrow$  FactorJEPA-RAW  $\rightarrow$  FactorJEPA separates gains due to parameter adaptation, predictor structure, and structured supervision. All methods share the same clips, masking policy, evaluation split, and metric implementations; we report per-encoder test results with 95% BCa bootstrap confidence intervals.

**FactorJEPA vs. every fine-tuning rival: the four predictive metrics it wins**  
(Stratified 10k · n<sub>test</sub> = 1,825 · 95% BCa CI)



ENCODER KEY · short code = full name · ↓ = lower is better (all four are error metrics)  
green shades = FactorJEPA / OURs · other colours = fine-tuning rivals  
DI = interaction factor D<sub>I</sub> · 3-stage = Layout → Agent → Interaction surgery

- 3sDI = FactorJEPA · 3-stage (L→A→interaction)
- 3sDI-diH = FactorJEPA · 3-stage (interaction-heavy)
- 3sDI-int = FactorJEPA · 3-stage + intervention
- raw = FactorJEPA-RAW (no factor targets)
- wSE-f30 = FactorJEPA + WiSE-FT merge 30%
- Auto-RGN = Auto-RGN gradient-norm surgery
- DoRA = DoRA (weight-decomposed LoRA)
- LP-FT = LP-FT (linear-probe then fine-tune)
- LoRA = LoRA (low-rank adaptation)
- cont-SSL = continual SSL (V-JEPA pretrain)
- frozen = frozen V-JEPA 2.1 encoder
- full-FT = full fine-tuning

Figure 6: *FactorJEPA outperforms every fine-tuning rival on the four predictive metrics it wins outright* (Future-frame L1 at 2B; Causal future-block L1, Mask-ratio slope, exposure-bias gap at 1B). Bars show held-out score ± 95% BCa CI; every rival is named in the key. Full 15-diagnostic scorecard in the supplementary material. wSE = WiSE-FT (weight-space ensembles for fine-tuning).

#### Four Diagnostics of Predictive Structure

We report four predictive-structure diagnostics (formal definitions in Appendix E): *Future-frame L1* (↓), the normalized distance between predicted and target future embeddings; *Causal future-block L1* (↓), a strictly causal test that forecasts the target encoder’s future-half latents from the past half alone; *Mask-ratio slope* (↓), the growth in prediction error as evidence is progressively masked; and *Motion cosine* (↑), intra- minus inter-class cosine similarity across motion

classes.

#### Decisive Gains over the Frozen World Model

Figure 5 reports FactorJEPA’s advantage over the frozen backbone in confidence-interval widths (values beyond 1× indicate separation): it clears the threshold on all four headline diagnostics at both scales, and stays decisive from 2B to 1B. At 1B, FactorJEPA cuts future-frame L1 by 10.6%, causal future-block L1 by 8.9%, and mask-ratio slope by

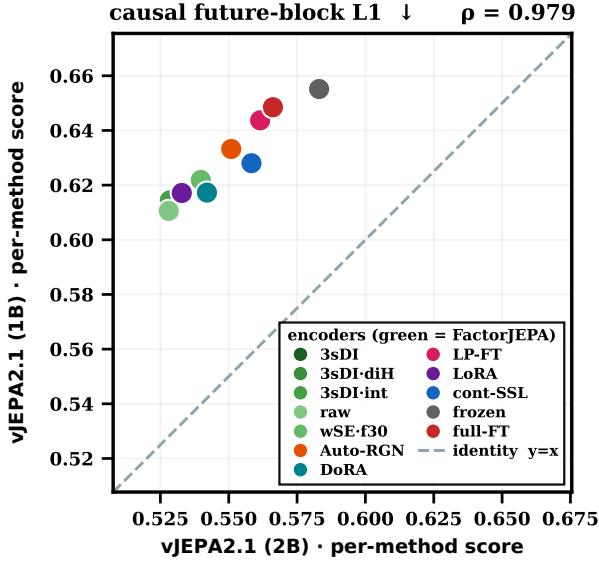


Figure 7: **Method rankings survive the 2B→1B backbone change.** Per-method *Causal future-block L1* at both scales (Spearman  $\rho = 0.978$ ), making the 1B backbone a low-cost proxy for 2B. Full 15-panel grid in the supplementary material.

27.4% against the frozen backbone, and lifts motion-cosine separation to  $15.3\times$  its frozen value, each gain exceeding  $70\times$  the paired 95% CI. Against the strongest *competing* adaptation (Figure 6) it further separates from *every* fine-tuning rival on Future-frame L1 and Causal future-block L1 (both scales), Mask-ratio slope, and free-running exposure-bias gap (1B), for 24 of 24 paired wins over the six baselines, and it beats the strongest full-scale rival, LoRA, on all four headline diagnostics (+2.8%, +1.6%, +25.2%, and +17.8%). The gain is structured, not universal: FactorJEPA leads the predictive and masking diagnostics while frozen and conventionally tuned models keep motion-linearity and some frame-timing (Misra, Zitnick, and Hebert 2016; Wang, Jiao, and Liu 2020) and temporal-coherence (Wei et al. 2018; Dwibedi et al. 2019) measures. The lone trade-off is Motion cosine, where factorization favours accurate, causally structured prediction over maximally linear motion readout (per-metric magnitudes in Appendix D).

### Cross-Scale Replication

The complete ranking of adaptation methods is stable across the 2B and 1B backbones (20 arms at 2B, 14 at 1B). The per-diagnostic Spearman correlations  $\rho_k$  are strongest on the four headline metrics ( $\rho = 0.895$  to  $0.978$ ; Figure 7), twelve of fifteen diagnostics retain consistent rankings, and only Temporal-order, Teacher-free gap, and Rollout drift fail to transfer (full comparison in the supplementary material). This establishes the 1B model as a faithful lower-cost proxy for 2B, and shows the gains and the motion trade-off are properties of the method ordering, not artifacts of one backbone.

### Ablations

We isolate each ingredient with matched controls on the same held-out split (paired BCa bootstrap; *separates* means the oriented 95% interval excludes zero); full per-metric detail is in Appendix D. **The gain is predictor surgery, not the factor targets:** FactorJEPA-RAW (identical schedule and unfreeze budget, no factor supervision) is not beaten by full FactorJEPA on future-frame at either scale, so the closing of the frozen-to-adapted gap comes from the surgical adaptation, not the factor-view supervision layered on top. **Updating the encoder, not only the head, drives the wins:** head-only adaptation loses on future-frame, causal future-block, and mask-ratio (each separated) but wins back motion-cosine, rollout, and exposure-bias. **The interaction stage adds no measurable value:** removing  $D_I$  (no- $D_I$ ) leaves every predictive diagnostic statistically unchanged. **Weight merging hurts:** a WiSE-FT (weight-space ensembles for fine-tuning) (Wortsman et al. 2022) interpolation (mixing fractions 0.3, 0.5, 0.7) loses to the un-merged model on future-frame and motion-cosine at both scales.

**Limitations.** Three controls bound our claims. A compute-matched control (plain pretraining for the same number of steps as the surgery) is declared but not run, so we cannot fully separate the surgical schedule from additional optimisation; head-only pretraining and the continual-learning baselines (CaSSLe (Fini et al. 2022), EWC (elastic weight consolidation) (Kirkpatrick et al. 2017)) were trained but not evaluated on the suite; and the method is a precondition on the base checkpoint, as on V-JEPA 2.0 the same recipe does not improve over the frozen encoder, so the gains are specific to V-JEPA 2.1. At the 10k screening scale factorization also concedes maximally linear motion readout to full fine-tuning (−44.9% at 2B, −33.4% at 1B), a trade-off that reverses only at full (86,831-video) scale.

### Conclusion

We studied world modeling in populous, crowded, and chaotic Global South cities, where heterogeneity, occlusion, and mixed-traffic negotiation challenge conventional JEPA predictors. We introduced DENSEWORLD: 276 hours of drive-through, walk-through, and aerial video collected across 22 cities. Building on this benchmark, we proposed FactorJEPA, which makes world structure a first-class predictive primitive through *predictor surgery*: a three-stage curriculum that adapts the predictor over automatically separated *layout*, *agent*, and *interaction* views. Across 2B and 1B backbones, FactorJEPA delivers (i) more accurate future-latent prediction, (ii) stronger causal future-block structure, and (iii) greater robustness under reduced visual evidence, with method rankings reproducing across scale at  $\rho = 0.895$  to  $0.978$ . The results also expose a trade-off between structured prediction and maximally linear motion readout at the 10k screening scale, which reverses at full scale (86,831 factor-annotated videos). Together, these contributions move JEPA world models from *monolithic predictors* toward *structured, factor-aware models* of complex real-world dynamics.

The DENSEWORLD dataset, the training and evaluation pipeline, the released checkpoints, and the complete 15-

diagnostic harness are made available under the CC BY 4.0 license.

## References

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